GWSPH RESEARCH SHOWCASE

BIOSTATISTICS AND BIOINFORMATICS

Mobile DNA Activity in Parkinson's Disease: A Locus-Specific View of Endogenous Retroviruses

Introduction: Human endogenous retroviruses (HERVs) are mobile genetic sequences derived from ancient retroviral infections. While typically silenced, their reactivation has been implicated in gene dysregulation, aging, and immune-related transcriptional pathogenesis of some neurodegenerative diseases. Parkinson's disease (PD) is the second most common neurodegenerative disorder, yet its etiology and HERV reactivation remain poorly understood. This study investigates locus-specific HERV expression, associated with early-stage PD that includes genetic and non-genetic cases (all PD), and idiopathic PD without a known genetic cause (iPD).

Methods: We analyzed RNA-seq whole-blood samples from 492 individuals (358 all PD, of which 256 were iPD, and 134 healthy controls (HC)). Locus-specific HERV expression was quantified using Telescope, followed by differential expression analysis with DESeq2. We additionally performed host gene expression analysis, gene set enrichment of HERV-proximal genes, and immune cell deconvolution using CIBERSORTx.

Results: Differential expression analysis identified 45 significantly dysregulated retrotransposon loci, comprising 20 HERVs in all PD versus HC. HERVH was the most recurrently dysregulated family, with 6 upregulated loci distributed across multiple chromosomes, while ERVLE demonstrated mixed regulation across three loci. Eleven loci overlapped between all PD and iPD analyses with consistent directional regulation, including HERVH and HERVL, supporting robust disease-associated retroviral signals independent of genetic subtype. Gene set enrichment analysis of HERV-proximal genes revealed suppression of tRNA metabolic processes, DNA repair, RNA processing, and mitochondrial organization, which are pathways central to PD pathogenesis, alongside activation of cell motility and chemotaxis. Cell deconvolution identified increased neutrophil abundance and decreased resting CD4+ memory T cell proportions, consistent with the chemotaxis activation found in all PD versus HC. Together, these results suggest that HERV dysregulation may contribute to systemic immune perturbation in PD.

Conclusion: Our analysis demonstrates that PD-related transcriptomic HERV alterations in whole blood are reproducible across PD populations and are associated with dysregulation of fundamental cellular pathways and peripheral immune remodeling. These findings suggest that locus-specific HERV expression may serve as a candidate blood-based signature of PD-associated neuroinflammatory and neurodegenerative processes.

Primary Presenter

Elizabeth Banda-Arnold

Co-Presenter(s)

Status

Masters

Authors

Elizabeth Banda-Arnold

Charles S. Venuto

Keith A. Crandall

Research Mentor/ Department Chair

Keith A. Crandall